Parallel lines and transversals

1

a Consecutive exterior angles

b Consecutive interior angles

c Alternate interior angles

d Corresponding angles

e Corresponding angles

f Alternate exterior angles

g Consecutive interior angles

h Alternate interior angles

i Consecutive exterior angles

j Alternate interior angles

k Corresponding angles

l Consecutive interior angles

2

a
- i Congruent
- ii They are corresponding angles.

b
- i Supplementary
- ii They are consecutive interior angles.

c
- i Congruent
- ii They are alternate interior angles.

3 $\angle APB$

4 $\angle DQF$

5 $\angle AQC$

6

a $x = 125$

b $x = 125$

c $x = 85$

d $x = 73$

e $x = 104$

f $x = 72$

7

a $p = 64$

b $q = 64$

8

a $g = 32$

b $f = 58$

9

a $x = 39$

b $x = 45.5$

c $x = 35$

10

a They are alternate interior angles.

b They are supplementary angles.

c 78°

11

a They are vertical angles.

b They are supplementary angles.

c 74°

12

a They are consecutive interior angles.

b They are supplementary angles.

c 102°

13

a $x = 114, y = 54$

b $x = 40, y = 50$

c $x = 21, y = 17$

d $x = 9, y = 22$

14 $\angle ADC$ and $\angle DCR$

15 $\angle NBP$ and $\angle DCR$

16

a 47

b 80

c 100

17

a b, f, h

b a, d

c g

Additional questions

18

a i Corresponding angles

ii The angles are congruent

iii $x = 125$

b i Alternate interior angles

ii The angles are congruent

iii $x = 75$

c i Consecutive interior angles

ii The angles are supplementary

iii $x = 55$

d i Consecutive exterior angles

ii The angles are supplementary

iii $x = 76$

e i Corresponding angles

ii The angles are congruent

iii $x = 107$

f i Alternate exterior angles

ii The angles are congruent

iii $x = 96$

19 $m\angle 1 = 51^\circ$ because it is a corresponding angle to the given angle. $\angle 1$ and $\angle 2$ are consecutive interior angles so they are supplementary. Using the definition of supplementary angles and solving for $\angle 2$,

$m\angle 2 = 180^\circ - 51^\circ$

$m\angle 2 = 129^\circ$

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a No. The Consecutive Exterior Angles Theorem states that the angles are supplementary. $m\angle 1 = 180^\circ - 84^\circ$ so $m\angle 1 = 96^\circ$

b No, alternate interior angles are congruent so $m\angle 1 = 104.75^\circ$

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a $x = 114$
$y = 54$

b $x = 40$
$y = 50$

c $x = 35$

d $x = 39$

e $x = 45.5$

f $x = 9$
$y = 22$

22

a Yes we can conclude $\angle 5 \cong \angle 9$. Since $a \parallel b$, $\angle 1 \cong \angle 5$ by the corresponding angles postulate. Since $\angle 1 \cong \angle 9$, $\angle 5 \cong \angle 9$.

b Corresponding angles are congruent when the transversal cuts across parallel lines. More information is needed here. The angles would not be congruent if the two lines

23 We are given that $\overleftrightarrow{GH} \parallel \overleftrightarrow{KL}$ so we know that $\angle 2 \cong \angle 6$ by the corresponding angles postulate. Then we have $\angle 6 \cong \angle 8$ by the vertical angles theorem. Therefore $\angle 2 \cong \angle 8$ by the transitive property of congruence.

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a It is given that $a \parallel b$. So $\angle 2$ and $\angle 3$ are supplementary by the consecutive interior angles theorem. It is also given that $c \parallel d$. So by the consecutive interior angles theorem, $\angle 1$ and $\angle 2$ are supplementary. Therefore, by the congruent supplements theorem,$\angle 1 \cong \angle 3$

b

To prove: $\angle 1 \cong \angle 3$

	Statements	Reasons
1.	$a \parallel b$	Given
2.	$\angle 1 \cong \angle 4$	Alternate interior angles theorem
3.	$c \parallel d$	Given
4.	$\angle 4 \cong \angle 3$	Corresponding angles postulate
5.	$\angle 1 \cong \angle 3$	Transitive property of congruence

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To prove: $\angle HBE \cong \angle BEJ$

	Statements	Reasons
1.	$\overleftrightarrow{AF} \parallel \overleftrightarrow{GL}$	Given
2.	$\angle BHJ$ and $\angle HBE$ are supplementary	Consecutive interior angles theorem
3.	$\angle HJE$ and $\angle BEJ$ are supplementary	Consecutive interior angles theorem
4.	$\angle BHJ \cong \angle HJE$	Given
5.	$\angle BHJ$ and $\angle BEJ$ are supplementary	Substitution
6.	$\angle HBE$ and $\angle BEJ$	Congruent supplements theorem